VIII Congreso bienal
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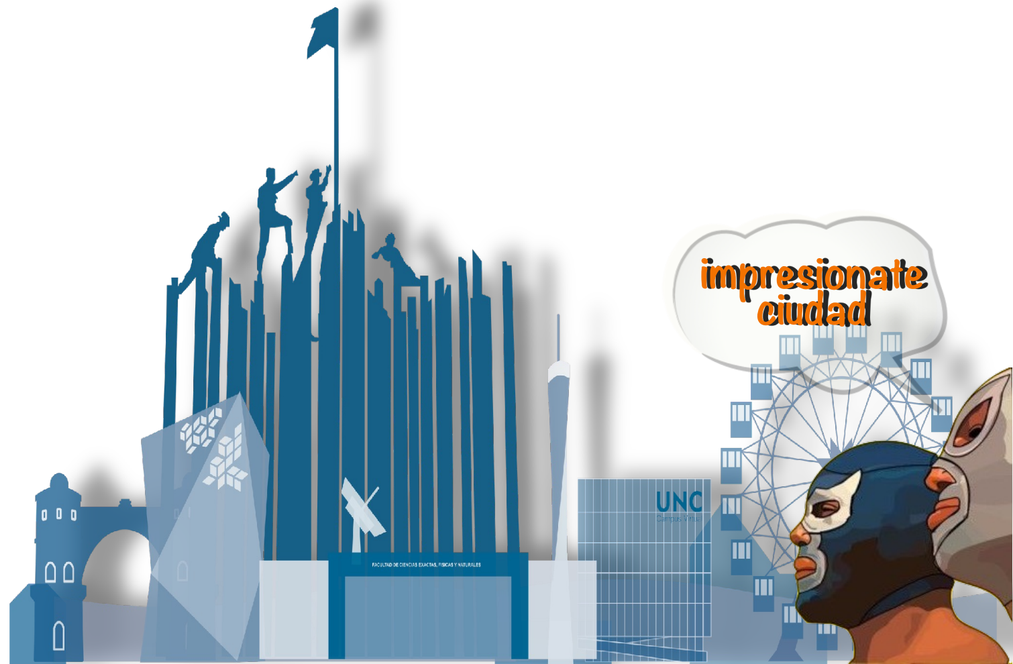
Parallel-Plate Waveguide Remote Excitation with Orthogonal and Parallel RF Coils for 15.2 T MRI

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Casa abierta al tiempo
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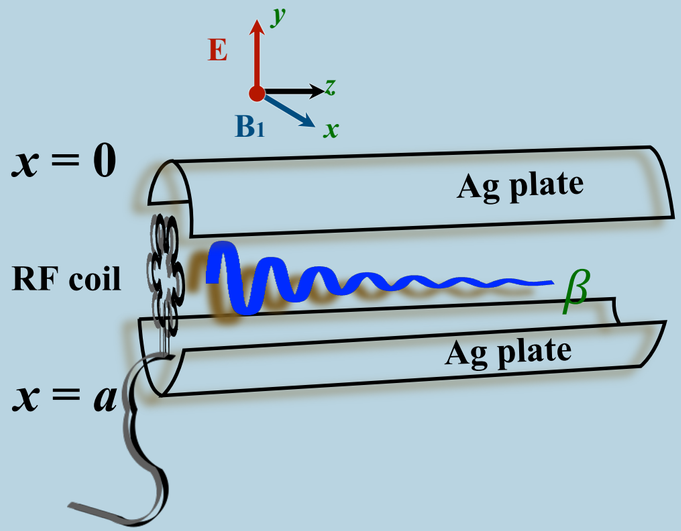
MAX-PLANCK-GESELLSCHAFT



MOTIVATION & OBJECTIVE

Challenge at Ultra-High Field (UHF) MRI:

- Higher B_0 → increased sensitivity, susceptibility contrast & chemical shift dispersion
- BUT: RF wavelength approaches sample dimensions → **B_1 inhomogeneities & reduced penetration depth**
- Traveling-wave MRI (twMRI) is a promising alternative, but conventional waveguides have a **critical frequency** dependent on bore dimensions



Objective: Investigate feasibility of remote RF excitation using a **Parallel-Plate Waveguide (PPWG)** at **15.2 T (650 MHz)** with two coil orientations (perpendicular & parallel to plates)

METHODS

Waveguide Prototype

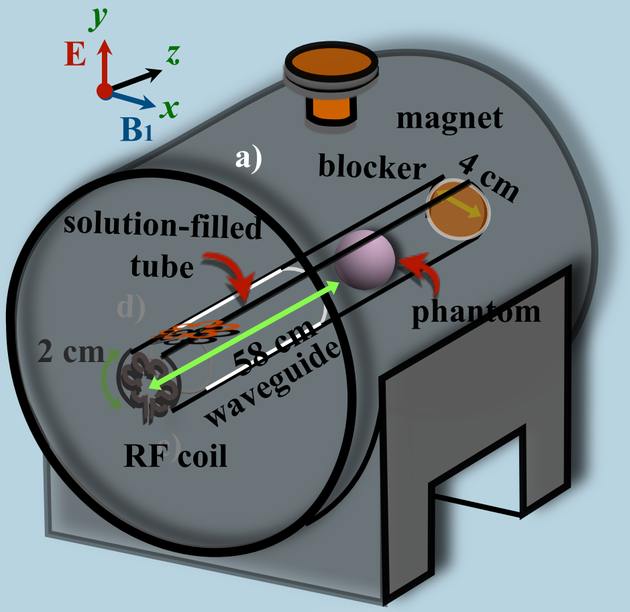
- **2 aluminum strips** (1 cm width, 6 μm thickness) on plastic tube (120 cm length, 2 cm radius)
- Copper blocker at one end to confine RF signal
- Saline-filled pipe (3 cm dia, 55 cm length, $\varepsilon_r \approx 80$) as dielectric medium

RF Coil

- **Flower-shaped surface coil** tuned to 650 MHz (bio-inspired design)
- Transceiver mode, linear drive
- Positioned **completely outside the magnet** → true remote excitation

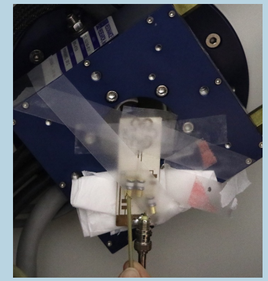
RF Characterization

- **$S_{11} \approx -33$ dB** at 650 MHz for BOTH orientations → excellent impedance matching





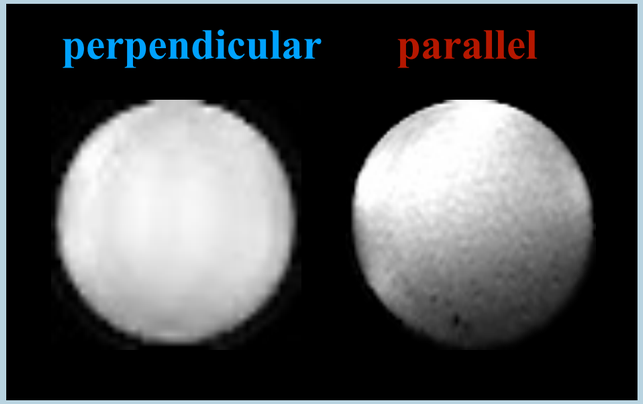




RESULTS

Phantom Imaging (Saline Sphere)

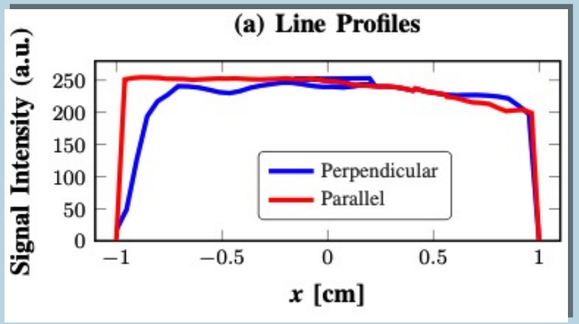
- Excellent image quality despite coil–phantom separation (~20× coil diameter)
- Perpendicular: higher central signal, Gaussian-like profile (peak 246 a.u., FWHM 0.85 cm)
- Parallel: more uniform plateau (250–254 a.u.), asymmetric decay

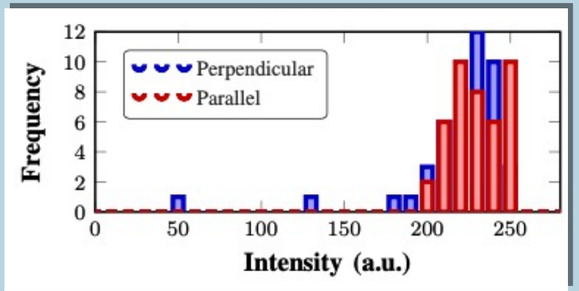


SNR Quantification (ACR standard deviation method)

Orientation	SNR
Perpendicular	42.9 ± 2.1
Parallel	33.3 ± 1.8

$$SNR_{Gain\%} = \frac{SNR_{per} - SNR_{par}}{SNR_{par}} 100 = 28\%$$



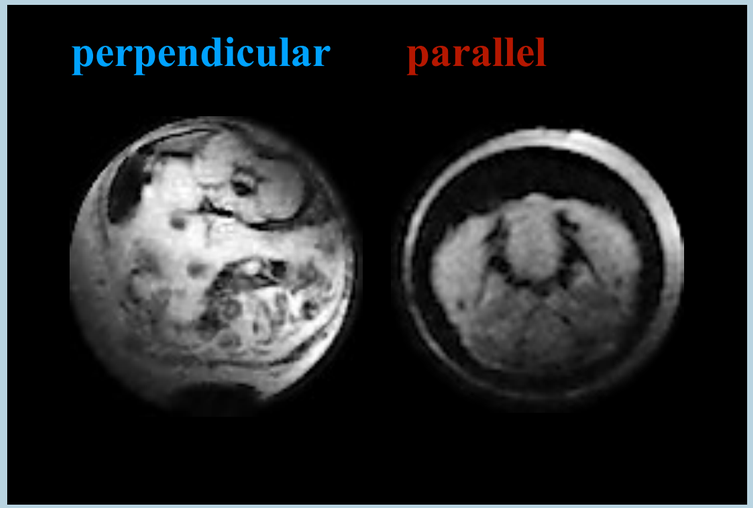


RESULTS

Ex Vivo Mouse Imaging

- Both orientations produce **clear anatomical detail**
- Brain: cerebral cortex, hippocampus, cerebellar folia identifiable
- Abdomen: liver, kidneys, intestinal structures delineated
- Image quality comparable to conventional RF coils







rat phantom

Key Advantages of PPWG

Advantage	Explanation
Frequency independence	TEM mode → zero cutoff at all frequencies
Scalability	Dimensions adaptable to any bore size
Cost-effectiveness	Simple materials (Al strips, plastic tube)
Orientation flexibility	Both coil orientations yield acceptable images
Space decoupling	External transmitter → unlimited coil size, more room in bore



CONCLUSION

- Remote excitation via surface coil + PPWG successfully transmits MR signals at **15.2 T**
- **Perpendicular orientation: 28% SNR gain** over parallel; both clinically acceptable
- Simple, low-cost, frequency-independent design compatible with standard pulse sequences
- Strong foundation for UHF MRI applications — scalable toward clinical human imaging

FUTURE DIRECTIONS

***in vivo* imaging** with physiological monitoring

Multi-nuclear imaging (^{13}C, ^{23}Na, ^{31}P) for metabolic studies

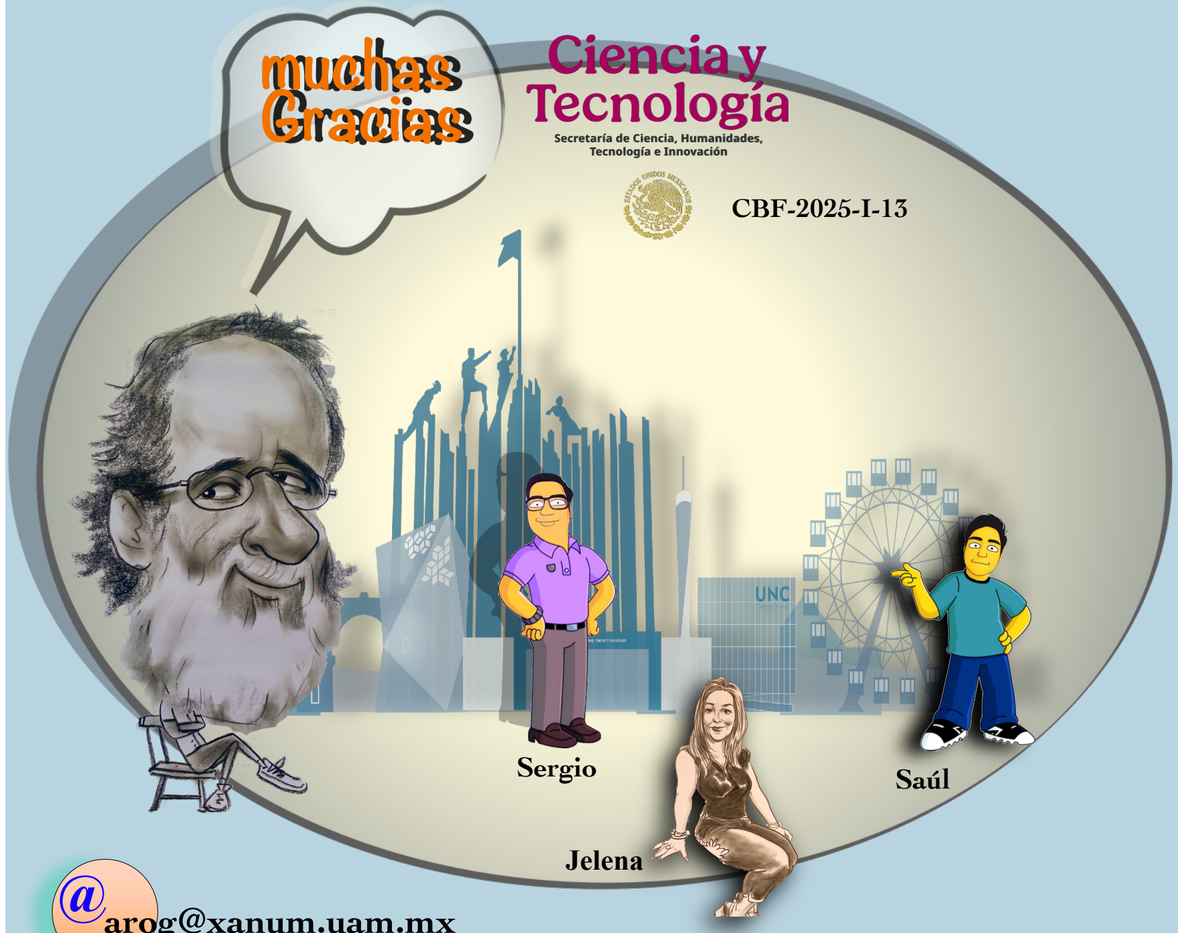
Functional MRI (fMRI) feasibility

Numerical simulations — field validation & SAR estimation

Human applications — scaling waveguide for clinical imaging

Coil design optimization, impedance matching, higher-conductivity materials, multi-coil arrays

Acknowledgments



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